# FILTERED RIGIDITY OF THE DEGREE-SEVEN JACOBIAN COUNTEREXAMPLE: A TRANSVERSE ALGEBRA OF LENGTH 584

NATHANIEL MONSON

ABSTRACT. Let $G\colon \mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$ be the normalized degree-seven counterexample to the Jacobian conjecture, and let $K_{3,7}$ be the finite-dimensional coefficient scheme of maps $H$ satisfying

$$\deg H \leq 7, \qquad H(0) = 0, \qquad JH(0) = I, \qquad \det JH = 1.$$

Unrestricted Keller deformations of $G$ are volume-preserving formal source reparametrizations. Ordinary degree is not preserved by those reparametrizations, and the bounded-degree scheme cuts the formal coordinate orbit nontransversely. After removing the normalized affine source action, an explicit torus-stable affine slice has a ten-dimensional tangent space, with weights

$$(-1, 2, -3, -2, -1, 0, 1, 1, 2, 3).$$

Nevertheless, its reduced completed local ring is $\mathbb{C}$.

We determine the entire nonreduced thickening by exact computer-assisted calculation. Its length is 584, its Hilbert–Samuel function is

$$(1, 10, 44, 108, 157, 145, 86, 30, 3),$$

and its maximal ideal satisfies $\mathfrak{m}^9 = 0 \neq \mathfrak{m}^8$. Equivalently, the normalized affine orbit is the reduced local component and occurs with transverse multiplicity 584. The Macaulay inverse system has 60 minimal contraction generators, while the Kuranishi ideal has 36 minimal equations: 11 quadratic, 13 cubic, 11 quartic, and one sextic. In particular, the local algebra has Cohen–Macaulay type 60 and is neither a complete intersection, Gorenstein, nor level. We state the exact specialization and rational-reconstruction certificates used in the proof and distinguish this bounded-degree, affine-quotiented theorem from unrestricted deformation claims.

*Working computer-assisted manuscript. The exact calculations have been rerun from the recovered source artifacts. A genuinely independent CAS implementation and specialist review remain submission requirements.*

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## 1. Introduction

Put $u = 1 + xy$, and define

$$P = u^3 z + y^2 u(4 + 3xy), \tag{1.1}$$

$$Q = y + 3xu^2 z + 3xy^2(4 + 3xy), \tag{1.2}$$

$$R = 2x - 3x^2 y - x^3 z. \tag{1.3}$$

The announced counterexample is $F = (P, Q, R)$; it has $\det JF = -2$ and is not injective [1, 5, 8]. We use the normalized map

$$G = (R/2, Q, P). \tag{1.4}$$

*Discovery attribution.* We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [8]. Then

$$G(0) = 0, \qquad JG(0) = I, \qquad \det JG = 1, \qquad \deg G = 7.$$

The disproof immediately raises a deformation question: is this map an isolated accident, or is it one point of a family? The question has no useful answer until both the allowed coordinate changes and a degree bound are specified. There are already unrestricted families of nonproper Keller maps [8, 2]. The theorem in this paper is instead local, bounded-degree, and transverse to a stated affine action.

Let $K_{3,7}$ be the coefficient scheme of polynomial maps $H \colon \mathbb{A}^3 \to \mathbb{A}^3$ of ordinary degree at most seven satisfying

$$H(0) = 0, \qquad JH(0) = I, \qquad \det JH = 1. \tag{1.5}$$

The normalized affine source action is

$$\rho_{a,A}(H)(x) = [JH(a)A]^{-1}\big(H(a + Ax) - H(a)\big), \qquad (a, A) \in \mathrm{Aff}_3\,. \tag{1.6}$$

We construct an explicit affine slice $S \subset K_{3,7}$ through $G$. Our main result describes

$$\mathcal{R} = \widehat{\mathcal{O}}_{S,G}.$$

The ring and all of its invariants may be formed over $\mathbb{Q}$; the statements below are unchanged after base extension to $\mathbb{C}$.

**Theorem 1.1** (Transverse local algebra)**.** *For the slice $S$ of definition 2.2, the following hold.*

*(i) The reduced completed local ring is a point:*

$$\mathcal{R}_{\mathrm{red}} \cong \mathbb{C}.$$

*(ii) If $\mathfrak{m}$ is the maximal ideal, then*

$$\operatorname{length} \mathcal{R} = 584, \qquad \mathfrak{m}^9 = 0 \neq \mathfrak{m}^8,$$

*and*

$$\left(\dim_{\mathbb{C}} \mathfrak{m}^d/\mathfrak{m}^{d+1}\right)_{d=0}^{8} = (1, 10, 44, 108, 157, 145, 86, 30, 3).$$

*(iii) The Macaulay inverse system of $\mathcal{R}$ has 60 minimal contraction generators. Their top divided-power degrees are distributed as*

| *degree* | 5 | 6 | 7 | 8 |
|---|---|---|---|---|
| *number* | 2 | 33 | 22 | 3. |

*Consequently* $\dim_{\mathbb{C}} \operatorname{Soc}(\mathcal{R}) = 60$.

*(iv) The Kuranishi ideal has 36 minimal generators, distributed by initial order as*

| *order* | 2 | 3 | 4 | 6 |
|---|---|---|---|---|
| *number* | 11 | 13 | 11 | 1. |

*For the minimal resolution over* $\mathbb{C}[[u_1, \dots, u_{10}]]$*, the extremal Betti numbers are* $\beta_1 = 36$ *and* $\beta_{10} = 60$.

The tangent space is not zero: it has dimension ten. Thus the content of theorem 1.1 is not infinitesimal rigidity. Every first-order transverse direction is obstructed, and all of the resulting infinitesimal thickness is captured by a comparatively large Artin algebra.

**Relation to concurrent work.** The mixed-sign grading of the map and its geometric significance were already public [4]. Teller computed a residual tangent dimension of ten and modular obstructions through finite order [6]. Mayner's repository contains an exact deformation calculation in a narrower fixed-direction, fixed-support ansatz; its provenance statement credits the mathematical work to Claude Fable 5 and the circulation and packaging to William G. P. Mayner [3]. These are important predecessors, but neither determines the reduced characteristic-zero germ of the full normalized degree-seven slice or its complete local algebra.

Our candidate contribution begins with the exact residual weight representation and continues through the characteristic-zero radical statement, length, inverse system, socle, and minimal equations. We make no claim to the tangent dimension alone, qualitative nonreducedness, the torus action itself, or rigidity in a restricted ansatz.

**Formal orbit versus bounded degree.** A Keller map is étale. Formal étaleness gives unique lifts, so every unrestricted normalized Keller deformation comes from a volume-preserving formal change of source coordinates [7]. This standard fact does not imply the present theorem. The formal changes need not preserve ordinary degree at most seven. The geometry below is the high-order incidence of that formal orbit with the bounded-degree coefficient space.

**Proposition 1.2** (Formal source-orbit theorem)**.** *Let $A$ be a local Artin $\mathbb{C}$-algebra, and let $H_A$ be a normalized Keller deformation of $G$ over $A$ reducing to $G$. There is a unique normalized formal $A$-automorphism of $\widehat{\mathbb{A}}^3_0$, $\phi_A \equiv \mathrm{id}$* $\pmod{\mathfrak{m}_A}$ *such that*

$$H_A = G \circ \phi_A.$$

*Moreover,* $\det J\phi_A = 1$*. Consequently the degree-*$\leq 7$ *deformation functor is the subfunctor of volume-preserving source reparametrizations cut out by*

$$\deg(G \circ \phi_A) \leq 7.$$

*Proof.* On the completed local source, formal étaleness makes $G_A$ a formal isomorphism. Set $\phi_A = G_A^{-1} \circ H_A$. It is the unique formal automorphism reducing to the identity and satisfying $G_A \circ \phi_A = H_A$. Finally,

$$1 = \det JH_A = \det JG(\phi_A) \det J\phi_A = \det J\phi_A.$$

The normalizations at the origin follow from uniqueness. □

## 2. The affine action and a transverse slice

The coefficient affine space underlying (1.5), before the determinant equations are imposed, has dimension

$$3\left(\binom{10}{3} - 4\right) = 348.$$

The map $G$ is equivariant for the source torus

$$A_\tau = \mathrm{diag}(\tau^{-1}, \tau, \tau^2), \qquad G(A_\tau x) = A_\tau G(x). \tag{2.1}$$

**Proposition 2.1** (Normalized affine stabilizer)**.** *The stabilizer of* $G$ *under* (1.6) *is*

$$\mathrm{Stab}_{\mathrm{Aff}_3}(G) = \{A_\tau : \tau \in \mathbb{G}_{\mathrm{m}}\} \cong \mathbb{G}_{\mathrm{m}}.$$

*Proof.* Suppose $\rho_{a,A}(G) = G$. With

$$\alpha(x) = a + Ax, \qquad \beta(v) = G(a) + JG(a)Av,$$

one has $G \circ \alpha = \beta \circ G$. The omitted locus of $G$ is

$$\Gamma = \left\{ \left(\frac{s}{3}, \frac{2}{s}, \frac{1}{3s^2}\right) : s \in \mathbb{C}^\times \right\} = V(3v_1v_2 - 2,\ 27v_1^2v_3 - 1) \cong \mathbb{G}_{\mathrm{m}}$$

[8]. Hence $\beta$ preserves $\Gamma$. Restrictions of affine-linear functions to this parametrization span

$$\left\langle 1, s, s^{-1}, s^{-2} \right\rangle \subset \mathbb{C}[s, s^{-1}].$$

The inversion $s \mapsto \mu/s$ does not preserve this space, whereas $s \mapsto \mu s$ does. No nonzero affine-linear function vanishes on $\Gamma$, so the ambient affine extension is unique and equals

$$\mathrm{diag}(\mu, \mu^{-1}, \mu^{-2}).$$

It remains to identify the source lift. The generic fiber is described by

$$cv^3 - 2v^2 + bv - 2a = 0.$$

This irreducible cubic has nonsquare discriminant

$$-4(27a^2c^2 - 18abc + 16a + b^3c - b^2);$$

indeed, the discriminant of the parenthesized expression as a quadratic in $a$ is $-4(3bc-4)^3$. The normal closure therefore has group $S_3$. For the corresponding subgroup $S_2 \subset S_3$,

$$N_{S_3}(S_2)/S_2 = 1,$$

so the cubic function-field extension has no nontrivial automorphism over the target field. The known source lift is $A_{\mu^{-1}}$, and uniqueness forces $\alpha = A_{\mu^{-1}}$. Finally, finite-type group schemes in characteristic zero are smooth, so no nonreduced stabilizer remains. □

**Definition 2.2** (The slice)**.** For a perturbation $H - G$, set the coefficients of the following eleven component–monomial pairs to zero:

| component | monomials |
|---|---|
| 1 | $y^2,\ xz,\ xy,\ x^2,\ x^3$ |
| 2 | $y^2,\ xz,\ xy,\ x^2$ |
| 3 | $y^2,\ xy.$ |

Let $A_{\mathrm{sl}}$ be the resulting 337-dimensional affine subspace and put

$$S = K_{3,7} \cap A_{\mathrm{sl}}.$$

**Proposition 2.3** (Transversality)**.** *The slice is torus-stable. The selected* $11 \times 11$ *affine-orbit minor has determinant* $-23328$*. In particular,*

$$\mathrm{Aff}_3 \times^{\mathbb{G}_\mathrm{m}} S \longrightarrow K_{3,7}$$

*is étale at* $[1, G]$*.*

*Proof.* The eleven coefficient perturbations are weight vectors for (2.1); the two whose base coefficients are nonzero have weight zero. This proves stability.

First work in the smooth 348-dimensional ambient coefficient space $\mathcal{A}$. The action map

$$\mathrm{Aff}_3 \times^{\mathbb{G}_\mathrm{m}} A_{\mathrm{sl}} \longrightarrow \mathcal{A}$$

has source and target of the same dimension. Differentiating (1.6) in the twelve affine parameters, quotienting the one-dimensional stabilizer, and applying the eleven displayed coefficient functionals gives determinant $-23328$. The ambient action map is therefore étale at $[1, G]$.

The Keller scheme $K_{3,7} \subset \mathcal{A}$ is invariant under the normalized affine action. Hence the inverse image of $K_{3,7}$ under the ambient action map is, near $[1, G]$,

$$\mathrm{Aff}_3 \times^{\mathbb{G}_\mathrm{m}} (K_{3,7} \cap A_{\mathrm{sl}}).$$

The claimed morphism is thus the base change of the ambient étale map along $K_{3,7} \hookrightarrow \mathcal{A}$, and is étale. □

**Corollary 2.4** (Local product and multiplicity)**.** *There is a noncanonical formal isomorphism*

$$\widehat{\mathcal{O}}_{K_{3,7},G} \cong \mathbb{C}[[t_1, \ldots, t_{11}]] \widehat{\otimes}_{\mathbb{C}} \mathcal{R}.$$

*Once* $\mathcal{R}_{\mathrm{red}} = \mathbb{C}$ *is proved below, the reduced local component is the normalized affine orbit and its scheme-theoretic transverse multiplicity is*

$$\operatorname{length} \mathcal{R} = 584.$$

*The formal local quotient stack is*

$$[\operatorname{Spf} \mathcal{R}/\mathbb{G}_{\mathrm{m}}].$$

*Proof.* The étale morphism of proposition 2.3 identifies the completed local germ with the product of the eleven-dimensional orbit direction and the slice, with the residual stabilizer $\mathbb{G}_{\mathrm{m}}$. Length of the transverse Artin factor is the component multiplicity. □

## 3. The equivariant Kuranishi map

Let

$$\Phi(H) = \det JH - 1.$$

The differential $D\Phi_G$ has rank 327 on the 337-dimensional slice. Choose tangent coordinates $u_1, \dots, u_{10}$ by taking the following free coefficient functionals; component indices in the table are one-based.

| | free coefficient | weight |
|---|---|---|
| $u_1$ | $H_3[x^2y^3]$ | $-1$ |
| $u_2$ | $H_3[xy^5]$ | 2 |
| $u_3$ | $H_3[x^4y^3]$ | $-3$ |
| $u_4$ | $H_3[x^4y^2z]$ | $-2$ |
| $u_5$ | $H_3[x^3y^4]$ | $-1$ |
| $u_6$ | $H_3[x^3y^3z]$ | 0 |
| $u_7$ | $H_3[x^3y^2z^2]$ | 1 |
| $u_8$ | $H_3[x^2y^5]$ | 1 |
| $u_9$ | $H_3[x^2y^4z]$ | 2 |
| $u_{10}$ | $H_3[xy^6]$ | 3. |

Formal implicit elimination of the 327 pivot coefficient directions gives a torus-equivariant Kuranishi map

$$\kappa\colon \widehat{\mathbb{A}}^{10} \longrightarrow \operatorname{coker}(D\Phi_G)$$

and an isomorphism

$$\mathcal{R} \cong \mathbb{C}[[u_1, \dots, u_{10}]]/I_\kappa. \tag{3.1}$$

The exact quadratic part spans an eleven-dimensional space. In particular, one equation begins with $(u_1 - 3u_5)^2$, but the quadratic cone is positive-dimensional; the reduced-rigidity theorem is not a consequence of the quadratic approximation.

## 4. Reduced local rigidity

We first isolate the geometric lemma that converts three smaller exact calculations into a statement about the full germ.

**Lemma 4.1** (Torus nullcone)**.** *Let $X$ be an affine finite-type $\mathbb{G}_{\mathrm{m}}$-scheme with a fixed point $0$, equivariantly embedded in*

$$V = V_- \oplus V_0 \oplus V_+.$$

*If the reduced germs at zero of*

$$X \cap V_-, \qquad X \cap V_0, \qquad X \cap V_+$$

*are all zero-dimensional, then the reduced germ $(X_{\mathrm{red}}, 0)$ is zero-dimensional.*

*Proof.* Suppose a positive-dimensional irreducible component $Y \subset X_{\mathrm{red}}$ passes through zero. The connected group $\mathbb{G}_{\mathrm{m}}$ preserves $Y$. If its action on $Y$ is trivial, then $Y \subset V_0$, a contradiction.

Otherwise the generic orbit has dimension one, and

$$\dim(Y/\!/\mathbb{G}_{\mathrm{m}}) \leq \dim Y - 1.$$

The fiber of $Y \to Y/\!/\mathbb{G}_{\mathrm{m}}$ over the image of zero therefore has positive local dimension at zero. Let $Z$ be a positive-dimensional irreducible component of that fiber through zero.

Every invariant regular function has on $Z$ its value at zero. Thus all weight-zero coordinate functions vanish on $Z$. If a positive-weight coordinate $x_i$ and a negative-weight coordinate $y_j$ were both nonzero at a point of $Z$, the balanced monomial

$$x_i^{|\operatorname{wt}(y_j)|} y_j^{\operatorname{wt}(x_i)}$$

would be a nonzero invariant there, a contradiction. Hence $Z \subset V_+ \cup V_-$. Irreducibility puts $Z$ in one of these two subspaces, contradicting the hypotheses. □

**Proposition 4.2** (Exact pure-weight certificates)**.** *For the Kuranishi germ of* (3.1)*, the following hold.*

*(i) On the positive locus, with parameters*

$$(p_1, p_2, p_3, p_4, p_5) = (u_7, u_8, u_2, u_9, u_{10}),$$

*the eliminated ideal contains*

$$p_1^6, \quad p_2^6, \quad p_3^3, \quad p_4^3, \quad p_5^2.$$

*(ii) On the negative locus, with parameters*

$$(n_1, n_2, n_3, n_4) = (u_1, u_5, u_4, u_3),$$

*the eliminated ideal contains*

$$n_1^4, \quad n_2^4, \quad n_3^4, \quad n_4^3.$$

(iii) *The weight-zero locus has one tangent parameter $q = u_6$. Its first nonzero compatibility equation occurs in order three and has coefficient*

$$\frac{212135552}{304438725}q^3.$$

*Computational proof.* The determinant equations are triangular by torus weight. On the positive locus, variables in the current weight block enter through $D\Phi_G$, while nonlinear terms use lower positive weights. Exact recursive elimination through weight six leaves five free tangent parameters; a rational Gröbner basis has 24 elements and contains the five displayed powers. The negative calculation reverses the weights. Elimination through absolute weight eight produces a 16-element rational Gröbner basis containing the four displayed powers.

In weight zero there are 21 coefficient variables and the linearized determinant map has rank 20. Solving the pivot variables formally in $q$ produces no order-two compatibility condition. At order three, the coefficient of $x^8y^4z^2$ is the displayed nonzero multiple of $q^3$. The scripts, exact outputs, and hash-pinned reproduction record are described in section 8. □

**Theorem 4.3** (Reduced affine rigidity)**.** *The reduced completed slice is a point:*

$$\mathcal{R}_{\mathrm{red}} \cong \mathbb{C}.$$

*Equivalently,*

$$\sqrt{I_\kappa} = (u_1, \ldots, u_{10}).$$

*Proof.* Apply lemma 4.1 to the torus-stable slice, using proposition 4.2. Each of the three pure-weight reduced germs is a point. Therefore the full reduced germ is zero-dimensional. Since its support contains the base point and the radical is a homogeneous maximal ideal in the completed tangent coordinates, the displayed equality follows. □

*Remark* 4.4. The theorem says that there is no nonconstant reduced branch through $G$ inside $K_{3,7}$, transverse to the normalized affine action. It does not say that $G$ has no degree-increasing deformation, no deformation before quotienting, or no deformation in a stabilized dimension.

## 5. Length, Hilbert function, and Loewy length

Write $S_0 = \mathbb{Q}[[u_1, \ldots, u_{10}]]$, let $R = S_0/I_\kappa$, and let $\mathfrak{m}$ denote its maximal ideal. The characteristic-zero computation is made over $\mathbb{Q}$; base change gives the complex ring of the preceding sections.

For $N \geq 0$, put

$$D_{\leq N} = \left(R/\mathfrak{m}^{N+1}\right)^\vee.$$

We use the divided-power convention, so multiplication by $u_i$ is dual to contraction

$$\partial_i Y^{[\alpha]} = Y^{[\alpha - e_i]}.$$

**Proposition 5.1** (The three terminal classes)**.** *The order-seven quotient has dimension* 581*. The degree-eight dual layer has dimension three, with one class in each torus weight* $-1, 0, 1$*. Consequently*

$$\dim_{\mathbb{Q}} R/\mathfrak{m}^9 = 584$$

*and*

$$\mathrm{ch}(\mathfrak{m}^8/\mathfrak{m}^9) = z^{-1} + 1 + z.$$

*Computer-assisted proof.* Take the good prime $p = 1{,}000{,}003$. The modular calculation gives

$$\dim R/\mathfrak{m}^9 \le 584,$$

with at most one new degree-eight class in each of the three displayed weights. Rank can only fall under this specialization. The modular quotient dimension is therefore an upper bound in characteristic zero.

For the reverse inequality, the exact Kuranishi recursion was evaluated on the coordinatewise down-set of the 60 degree-eight tangent monomials that can occur in the candidate classes. This produced 7,300 exact coefficients through order eight, including 990 in order eight. Every coefficient through order seven agrees with the full rational calculation, and all 990 order-eight coefficients agree modulo $p$ with the independently generated full modular output.

Chinese remaindering and rational reconstruction then give three rational dual classes. They have supports of sizes 192, 387, and 307 in weights $-1, 0, 1$, and satisfy respectively 3,045, 6,236, and 5,363 exact interacting rows. Each is normalized by a degree-eight coefficient equal to one, and their images are independent modulo $D_{\le 7}$. Thus the lower bound is $581+3 = 584$, proving equality. □

**Proposition 5.2** (Order-nine integrability certificate)**.** *One has*

$$\mathfrak{m}^9 = 0.$$

*Proof.* Let $\Lambda^{(9)}$ be a hypothetical nonzero top-degree-nine dual class of torus weight $W$. Every contraction $\partial_i \Lambda^{(9)}$ lies in the three-dimensional top degree-eight dual space. Hence

$$\partial_i \Lambda^{(9)} = a_i \Lambda^{(8)}_{W-w_i}$$

when $W - w_i \in \{-1, 0, 1\}$, and it is zero otherwise. Equality of mixed contractions produces a homogeneous linear system in the active scalars $a_i$.

For each possible weight $W = -4, \ldots, 4$, an exact square subsystem has the following determinant.

| $W$ | active scalars | determinant |
|---|---|---|
| $-4$ | 1 | 248832 |
| $-3$ | 2 | 23219011584 |
| $-2$ | 4 | 1597399996853114634 24 |
| $-1$ | 4 | 59902499881991798784 |
| 0 | 5 | 4416491511299491340746752 |
| 1 | 5 | $-1490565885063578327502028 8$ |
| 2 | 5 | $-16561843167373092517800 32$ |
| 3 | 3 | $-71328803586048$ |
| 4 | 1 | $-27648$ |

All are nonzero. For $|W| > 4$, there are no active contractions. Thus every $a_i$ is zero and all contractions of $\Lambda^{(9)}$ vanish, which forces $\Lambda^{(9)} = 0$. Therefore $\mathfrak{m}^9/\mathfrak{m}^{10} = 0$. Nakayama's lemma applied to the finitely generated module $\mathfrak{m}^9$ gives $\mathfrak{m}^9 = 0$. $\square$

**Corollary 5.3** (Hilbert–Samuel function)**.** *The ring $R$ has length* 584*, Loewy length nine, and Hilbert–Samuel function*

$$(1, 10, 44, 108, 157, 145, 86, 30, 3).$$

*Proof.* The exact ranks of the successive filtered quotients through degree seven are

$$(1, 10, 44, 108, 157, 145, 86, 30).$$

Proposition 5.1 adds the terminal dimension three, and proposition 5.2 proves there is no further layer. The sum is 584. $\square$

## 6. The complete inverse system

Let

$$D = I_\kappa^\perp \subset \mathbb{Q}_{\mathrm{DP}}[Y_1, \ldots, Y_{10}]$$

be the Macaulay inverse system. It is a contraction-stable $S_0$-module of dimension 584.

**Theorem 6.1** (Minimal inverse-system generators)**.** *The quotient $D/\mathfrak{m}D$ has dimension* 60*. Its top-degree and torus-refined generating polynomial is*

$$\begin{aligned}\mathcal{S}(t, z) = t^5&(z^{-1} + z^3)\\ &+ t^6(z^{-4} + z^{-3} + 5z^{-2} + 11z^{-1} + 8 + 6z + z^2)\\ &+ t^7(z^{-4} + 2z^{-3} + 3z^{-2} + 7z^{-1} + 3 + 3z + 2z^2 + z^3)\\ &+ t^8(z^{-1} + 1 + z).\end{aligned}$$

*Computer-assisted proof.* Modulo $p = 1{,}000{,}003$, the weight-block dimensions of $D$, of $\mathfrak{m}D$, and of their quotient are as follows:

| weight | $-7$ | $-6$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ |
|---|---|---|---|---|---|---|---|
| $\dim D_w$ | 1 | 4 | 12 | 28 | 47 | 74 | 94 |
| $\dim(\mathfrak{m}D)_w$ | 1 | 4 | 12 | 26 | 44 | 66 | 74 |
| dim quotient | 0 | 0 | 0 | 2 | 3 | 8 | 20 |

| weight | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\dim D_w$ | 96 | 86 | 67 | 45 | 23 | 7 |
| $\dim(\mathfrak{m}D)_w$ | 84 | 76 | 64 | 43 | 23 | 7 |
| dim quotient | 12 | 10 | 3 | 2 | 0 | 0. |

The quotient dimensions sum to 60, so specialization gives $\dim_{\mathbb{Q}} D/\mathfrak{m}D \leq 60$.

The certificate contains 60 exact rational divided-power polynomials with the filtered character displayed in the theorem. Their reductions form a basis of the modular quotient, so their characteristic-zero classes are independent and give the reverse inequality. Their complete contraction closure modulo $p$ has rank 584: 60 independent starting vectors produce 465 new independent first contractions and 59 new independent second contractions. Nakayama's lemma therefore proves that the 60 rational elements generate $D$. □

**Corollary 6.2** (Socle)**.** *The Cohen–Macaulay type is*

$$\dim_{\mathbb{Q}} \operatorname{Soc}(R) = 60.$$

*The associated-graded socle dimensions occur as*

$$\dim \frac{\operatorname{Soc}(R) \cap \mathfrak{m}^d}{\operatorname{Soc}(R) \cap \mathfrak{m}^{d+1}} = \begin{cases} 2, & d = 5, \\ 33, & d = 6, \\ 22, & d = 7, \\ 3, & d = 8, \\ 0, & \textit{otherwise.} \end{cases}$$

*In particular, $R$ is neither Gorenstein nor level.*

*Proof.* Macaulay duality gives a perfect pairing

$$\operatorname{Soc}(R) \times D/\mathfrak{m}D \longrightarrow \mathbb{Q}.$$

The filtered statement follows from

$$F_d D = (\mathfrak{m}^{d+1})^{\perp}$$

and the top-degree distribution in theorem 6.1. Type 60 excludes Gorensteinness, while the occurrence of socle classes in four distinct $\mathfrak{m}$-adic degrees excludes levelness. □

## 7. Minimal Kuranishi equations

**Theorem 7.1** (Thirty-six minimal equations)**.** *The conormal space of the Kuranishi ideal has dimension*

$$\mu(I_\kappa) = \dim_{\mathbb{Q}} I_\kappa/\mathfrak{m}I_\kappa = 36.$$

*Its initial-order distribution is*

$$11,\ 13,\ 11,\ 0,\ 1$$

*in orders* $2, 3, 4, 5, 6$*, respectively. The corresponding torus characters are*

$$E_2(z) = z^{-6} + z^{-5} + z^{-2} + z + z^2 + z^3 + 2z^4 + 2z^5 + z^6,$$
$$E_3(z) = z^{-3} + 2z^{-2} + 2z^{-1} + 1 + 2z + z^2 + z^3 + z^5 + 2z^6,$$
$$E_4(z) = z^{-8} + z^{-7} + 2z^{-6} + 2z^{-5} + z^{-4} + z^{-3} + z^{-2} + 2z^5,$$
$$E_6(z) = z^3.$$

*No new minimal generator first appears in orders seven, eight, or nine.*

*Computer-assisted proof.* Exact filtered Kuranishi elimination produces 36 independent classes with the displayed orders and weights. Independently within the computational pipeline, the Koszul complex of the modular algebra gives

$$\dim_{\mathbb{F}_p} H_1(u_1, \ldots, u_{10}; R_{\mathbb{F}_p}) = 36$$

with the same weight character. Since

$$H_1(u; R) \cong \operatorname{Tor}_1^{S_0}(R, \mathbb{Q}) \cong I_\kappa/\mathfrak{m}I_\kappa,$$

specialization gives the matching characteristic-zero upper bound. The 36 exact classes give the lower bound. □

**Corollary 7.2** (Extremal Betti numbers)**.** *For the minimal $S_0$-free resolution of $R$,*

$$\beta_1^{S_0}(R) = 36, \qquad \beta_{10}^{S_0}(R) = 60.$$

*The ring is not a complete intersection.*

*Proof.* The first equality is theorem 7.1. Since $R$ is Artinian and $S_0$ is regular local of dimension ten, local duality identifies the last Betti number with the type, which is 60 by corollary 6.2. A zero-dimensional complete intersection in a ten-dimensional regular local ring has ten minimal defining equations, not 36. □

## 8. Exact computation and reproducibility

All defining coefficients are rational. No floating-point arithmetic is used. The proof is divided into small exact certificates rather than one uninspectable final assertion.

The certificate layers are:

(1) stabilizer and residual-parameter identities, including the torus (2.1) and the omitted curve;

(2) the slice rank, ten tangent vectors and weights, and the quadratic obstruction space;
(3) the positive, negative, and fixed-locus calculations of proposition 4.2;
(4) 7,300 targeted rational Kuranishi coefficients through order eight and comparison of all 990 new coefficients with the full modular output;
(5) fresh CRT reconstruction and exact all-row checks for the three terminal dual classes;
(6) the nine order-nine determinants in proposition 5.2;
(7) the contraction quotient and closure, giving 60 minimal inverse-system generators and rank 584; and
(8) the Koszul $H_1$ calculation giving 36 minimal equations with their weight character.

The fresh audit reran the original SHA-256 manifests before executing the programs. Generated reduced-rigidity files agreed byte-for-byte with the recovered outputs. The order-eight classes were reconstructed afresh; the three runs required 17, 24, and 31 small primes and reproduced the archived rational supports and coefficients. The order-nine JSON and text certificates also agreed byte-for-byte. Finally, the three self-contained inverse-system verifiers reproduced type 60, closure rank 584, and Koszul $H_1$ dimension 36.

**What this verification does not establish.** The several checks are algebraically distinct, but they are not a genuinely independent implementation: the principal calculation and its verifiers belong to one Python/SymPy-centered family and share some exported data. They verify exact identities, ranks, specializations, and reconstructions. They do not independently validate the geometric choice of moduli problem, the interpretation of the slice, or the novelty claim.

Before release, the intended supplement will contain the exact Kuranishi rows, the 60 rational inverse-system generators, every verifier, dependency versions, source hashes, and immutable fresh logs. A second implementation in Singular or Macaulay2 remains a release gate.

## 9. Interpretation and open directions

The reduced answer is simple and the scheme-theoretic answer is not. The degree-seven counterexample is isolated in the stated reduced affine-moduli germ, yet it sits on a length-584 infinitesimal thickening. The ring has embedding dimension ten, 36 minimal equations, type 60, and socle in four different orders. Thus it is not well modeled by a hypersurface, complete intersection, Gorenstein algebra, or even a level algebra.

The number 584 does not count nearby maps. It is the intersection multiplicity created when the formally smooth source-reparametrization orbit meets the degree-seven coefficient stage. A one-variable model is

$$V(y - x^9) \cap V(y) = \operatorname{Spec} \mathbb{C}[[x]]/(x^9)\ :$$

the reduced intersection is one point, although the two conditions have ninth-order contact. The algebra $\mathcal{R}$ is a ten-variable, non-complete-intersection analogue. Its Hilbert function records the successive layers of the transverse normal cone, not a collection of distinct counterexamples.

Three questions are deliberately left outside the theorem.

**Question 9.1.** *Can the complete local algebra be recovered conceptually from the marked-root geometry, rather than by coefficient elimination?*

**Question 9.2.** *What is the reduced deformation germ when ordinary degree eight is allowed? Existing calculations give a mixture of exact and modular finite-order evidence, but not a characteristic-zero radical theorem.*

**Question 9.3.** *How does the Artin algebra change under polynomial, rather than affine, left–right equivalence or after stabilization?*

**Question 9.4** (Independent source-flow verification)**.** *Can one derive the finite overflow and Kuranishi equations directly from volume-preserving flows on the source? We ask for an implementation independent of the present coefficient-elimination code. Agreement would test the source-orbit interpretation of the local Artin algebra.*

## Appendix A. The first degree-eight families

The degree-seven theorem in the body is sharp as a statement about reduced affine rigidity. This appendix records what happens when the degree bound is raised by one. The explicit family is an exact theorem. The final residual calculation is reported with its mixed exact and modular evidentiary status.

A.1. **A three-parameter shear family.** For a polynomial $f \in \mathbb{C}[x, y]$, let

$$\phi_f(x, y, z) = (x, y, z + f(x, y)) \qquad \text{and} \qquad G_f = G \circ \phi_f.$$

The map $\phi_f$ is a polynomial automorphism with Jacobian determinant one. Hence every $G_f$ is a Keller map and has the same generic degree and the same failure of injectivity as $G$.

**Proposition A.1** (Quadratic shear family)**.** *If $f$ is a nonzero homogeneous quadratic, then $G_f$ has ordinary degree eight. More explicitly,*

$$G_f = G + f\left(-\frac{x^3}{2},\, 3x(1 + xy)^2,\, (1 + xy)^3\right),$$

*and its degree-eight homogeneous part is*

$$(0, 0, x^3y^3f(x, y)).$$

*In particular, $G_f$ is not affinely equivalent to the degree-seven map $G$.*

*Proof.* Only the variable $z$ is changed. Substitution into the three formulas (1.1)–(1.3), followed by the normalization (1.4), gives the displayed identity. If $f$ is quadratic, the only term of degree eight is $x^3y^3f$ in the final coordinate. Affine left–right equivalence preserves ordinary degree. $\square$

The stabilizer torus (2.1) acts on the space

$$\mathrm{Sym}^2\langle x, y\rangle = \left\langle x^2, xy, y^2\right\rangle$$

with weights $4, 2, 0$ after the common normalization induced by the $z$-shear. The following statement classifies the displayed family; it does not classify the full degree-eight Keller germ.

**Theorem A.2** (Affine classification inside the shear family)**.** *For nonzero homogeneous quadratics $f, g$,*

$$G_f \sim_{\mathrm{aff}} G_g \quad\Longleftrightarrow\quad g(x,y) = \tau^2 f(\tau x, \tau^{-1}y) \quad \textit{for some } \tau \in \mathbb{C}^\times.$$

*Consequently the generic affine quotient of the three-parameter quadratic shear family has dimension two.*

*Proof.* The displayed change of $f$ is induced by the stabilizer (2.1), so it gives affine equivalence. Conversely, an affine equivalence between two members preserves the degree-seven truncation. The normalized affine stabilizer of that truncation is $\{A_\tau\}$ by proposition 2.1. Comparing the degree-eight parts then gives exactly the displayed action on $f$. $\square$

A.2. **What remains after known shears.** The unrestricted formally étale lifting principle does not disappear in degree eight: it produces many formal coordinate directions. After removing the affine orbit and the explicit source and target shear directions, the recovered exact calculation leaves a 28-variable residual Kuranishi problem. Its only tangent weights are $-2$ and $-1$.

**Theorem A.3** (Transverse first-normal obstruction)**.** *In the stated residual slice, no additional formal branch has a first nonzero normal term transverse to the affine orbit and the known quadratic source- and target-shear components.*

*Exact computer-assisted proof.* Only weights $-2$ and $-1$ occur in the residual tangent space. In weight $-2$, exact rational elimination yields at successive orders the incompatible conditions

$$u_0u_1 = 0, \qquad 100u_0u_1 + 9 = 0.$$

Thus no nonzero weight-$-2$ initial direction lifts.

In weight $-1$, the order-four equations leave one projective line of directions. The order-five equations reduce it to one rational direction and one conjugate pair over $\mathbb{Q}(\sqrt{-2})$. Exact elimination rejects the rational direction at order six and the conjugate pair at order seven. These cases exhaust the projectivized weight-$-1$ initial directions. Hence every possible first transverse normal term is obstructed. $\square$

**Question A.4.** *Is the radical of the completed residual ideal exactly the union of the known source- and target-shear components? The transverse theorem excludes a branch with a first nonzero normal term, but does not by itself prove this scheme-theoretic radical equality or exclude a branch tangent to the known components to every computed order before leaving them.*

*Remark* A.5 (Scope). Proposition A.1 and theorem A.2 are conventional exact statements, and theorem A.3 is an exact computer-assisted statement in the named slice. No stronger scheme-theoretic description of the residual germ is asserted.

## Appendix B. An exact border presentation and finite-field resolutions

Let

$$S = \mathbb{Q}[[u_1, \dots, u_{10}]], \qquad R = S/I_\kappa$$

be the completed transverse local algebra constructed in the body. Its length, nilpotence order, and Hilbert–Samuel function are

$$\operatorname{length} R = 584, \qquad \mathfrak{m}^9 = 0 \neq \mathfrak{m}^8,$$

$$(1, 10, 44, 108, 157, 145, 86, 30, 3).$$

The calculations below replace the truncated expansion by a finite exact presentation.

**Theorem B.1** (Exact rational border basis)**.** *There is an explicitly recorded monomial order ideal $\mathcal{B}$ of cardinality* 584 *whose residue classes form a $\mathbb{Q}$-basis of $R$. Its degree distribution is the displayed Hilbert–Samuel function, and its torus-weight distribution is*

| $w$ | $-7$ | $-6$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\#\mathcal{B}_w$ | 1 | 4 | 12 | 28 | 47 | 74 | 94 | 96 | 86 | 67 | 45 | 23 | 7. |

*The corresponding border has* 2654 *monomials. Its exact rational relations generate $I_\kappa$. Equivalently, the ten multiplication matrices*

$$M_i \in \operatorname{Mat}_{584}(\mathbb{Q})$$

*commute, contain* 24402 *nonzero entries in total, preserve the weight and $\mathfrak{m}$-adic filtrations, annihilate degree eight after one further multiplication, and make the class of* 1 *cyclic.*

*Exact certificate proof.* The matrices were reconstructed from reductions at 32 primes near $10^9$; their entries had stabilized after 24 primes. The verification over $\mathbb{Q}$ does not rely on that stabilization. The 60 exact inverse-system generators and all their contractions give 584 linearly independent rational functionals. Pairing all 4082 variable-times-basis reductions against all 584 functionals gives

$$4082 \cdot 584 = 2383888$$

exact zero pairings.

Let $J$ be the ideal generated by the recorded border relations. The commuting multiplication matrices reduce every monomial to the span of $\mathcal{B}$, so $\dim_{\mathbb{Q}} S/J \leq 584$. The annihilation calculation gives $J \subseteq I_\kappa$, hence

$$S/J \twoheadrightarrow S/I_\kappa$$

and $\dim_{\mathbb{Q}} S/J \geq 584$. Therefore $J = I_\kappa$, and $\mathcal{B}$ is the asserted basis. □

Among the 5840 products $u_i b$, 1758 remain in $\mathcal{B}$; 4082 are border products, of which 1847 reduce to zero. The largest nonzero right-hand side has 84 terms. These figures are not inputs to the proof, but make the certificate independently implementable.

**Theorem B.2** (Three exact finite-field Betti tables)**.** *For each*

$$p \in \{1000003,\ 1000033,\ 1000001011\},$$

*all* 234 *nonzero torus-weight blocks of the Koszul complex give the same minimal-resolution Betti vector:*

$$\boxed{(1, 36, 354, 1565, 3937, 6216, 6432, 4403, 1942, 506, 60).}$$

*The character-valued Euler identity holds in every case.*

*Proof.* Reduce the exact border matrices modulo $p$, form the Koszul differential, decompose it by torus weight, and perform exact sparse elimination in each of the 234 nonzero blocks. The archived tables record every block rank and homology dimension. Recombining the blocks gives the displayed vector, and direct character arithmetic verifies

$$\sum_{i=0}^{10} (-1)^i \operatorname{ch} \operatorname{Tor}_i(R_p, \mathbb{F}_p) = \operatorname{ch}(R_p) \prod_{j=1}^{10} (1 - z^{w_j}).$$

□

*Remark* B.3 (Characteristic-zero boundary). The complete displayed vector is proved at the three named primes. Over $\mathbb{Q}$, the endpoint values

$$\beta_0 = 1, \qquad \beta_1 = 36, \qquad \beta_{10} = 60$$

are exact, while upper semicontinuity gives the displayed finite-field values as termwise upper bounds for the intermediate Betti numbers. The equality of all 234 blocks at three primes is strong evidence for the same rational table, but exact rational rank certificates for $\beta_2, \ldots, \beta_9$ remain an open computation.

**Question B.4** (Rational Betti certification)**.** *Produce exact rational rank or homology certificates for the intermediate Koszul blocks. The border presentation reduces this to a finite weight-by-weight calculation independent of the original Kuranishi elimination.*

## AI and computational disclosure

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. The mathematical evidence for the computer-assisted theorems consists of the exact certificates and reproducible calculations described above, not the output of a language model. Human specialist review has not yet been completed.

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